Campaign-Shift-Aware Social Bot Detection with Temporal Coordination Modeling and Selective Prediction

Author names and affiliations omitted

Abstract

Social bot detectors are commonly evaluated by randomly splitting accounts, although coordinated accounts may share campaign-specific behavior and therefore violate the practical assumption that deployment data resemble a random sample of the training mixture. This paper develops a campaign-aware evaluation and selective-prediction framework for studying detection reliability under campaign shift. The detector combines account-level profile and activity-rhythm features with a timestamp-aware co-activity graph. Candidate edges connect accounts that act on the same observable object within a temporal neighborhood; a parameterized mixture-of-exponentials kernel weights temporal proximity, while a time-randomization null estimates co-activity expected under a specified reference process. A graph neural network combines individual and relational evidence. For uncertainty, we evaluate bounded campaign-level prediction-set miscoverage using campaign-unit calibration. Because exchangeability across calibration and future campaigns cannot be presumed under campaign shift, the primary analysis treats this quantity empirically; a formal expected-risk guarantee is reported only if its assumptions and finite-sample requirements are independently justified. Neither marginal set coverage nor campaign-level miscoverage implies low error among automatically labeled accounts or robustness to arbitrary distribution shift. Leave-campaign-out cross-fitting is treated as a separate empirical evaluation strategy and is not assumed to inherit a fixed-model conformal guarantee. The experimental protocol compares random, campaign-disjoint, chronological and cross-dataset evaluation; includes graph, temporal and group-aware baselines; and reports detection, coverage, selective utility and computational cost. The manuscript specifies the evaluation protocol; empirical conclusions are reserved for experiments executed on audited datasets.

Keywords: Social bot detection; campaign shift; temporal coordination; co-activity graph; conformal risk control; selective prediction

1 Introduction

Social bots have become a persistent challenge for online information ecosystems because they can amplify content, manipulate engagement signals, and participate in coordinated information operations. Their detection has consequently evolved from conventional account-level classification based on profile attributes toward approaches that incorporate temporal behavior, social relationships, interaction patterns, and graph-based representations. Machine learning classifiers can exploit account metadata and activity statistics, while graph neural networks (GNNs) enable the integration of relational evidence from interconnected accounts. Temporal representation learning further provides mechanisms for modeling the evolution of interactions and behavioral patterns. Despite these developments, the effectiveness of a social bot detector depends not only on its architecture and input features but also on whether its evaluation protocol reflects the conditions encountered during deployment.

A particularly important challenge arises when the accounts encountered during deployment belong to campaigns that were not represented during model development. Accounts participating in the same campaign may exhibit similar activity rhythms, repeatedly interact with common objects, share profile characteristics, or coordinate their actions within narrow temporal intervals. When accounts are randomly divided into training and test sets, members of the same campaign may appear in both partitions. A detector can consequently exploit campaign-specific regularities that are shared across the partitions, producing an optimistic estimate of its ability to generalize to previously unseen campaigns. This problem is not resolved simply by adopting a more expressive neural architecture: a model may achieve strong random-split performance while remaining sensitive to changes in campaign behavior, coordination strategies, or data-collection conditions.

Campaign-aware evaluation therefore requires a distinction between several forms of generalization. Random account-level evaluation measures performance on held-out accounts drawn from a potentially familiar mixture of campaigns. Campaign-disjoint evaluation instead holds out complete, independently identified campaigns or eligible collection units. Chronological evaluation examines performance on observations occurring after a specified temporal cutoff, whereas cross-dataset evaluation assesses transfer between separately collected corpora. These settings represent different sources of distributional variation and cannot be treated as equivalent evidence of robustness. In particular, chronological separation does not necessarily eliminate campaign overlap, and campaign-disjoint evaluation does not by itself establish generalization across platforms. A rigorous evaluation must therefore measure these settings separately and control information leakage across training, validation, calibration, and test partitions.

Beyond generalization, reliable deployment requires the detector to recognize when its predictions may be insufficiently supported by the available evidence. Conventional classifiers usually produce a class label or probability estimate even for accounts exhibiting unfamiliar behavior or belonging to previously unseen campaigns. Selective prediction provides an alternative by permitting automatic decisions for selected accounts while deferring uncertain cases for additional analysis or human review. However, predictive discrimination, probability calibration, prediction-set coverage, and error among automatically classified accounts are distinct properties. High classification accuracy does not establish reliable uncertainty estimation, and high empirical coverage can be obtained by returning large prediction sets that provide limited decision utility. Moreover, marginal coverage over accounts does not establish low error within every campaign or guarantee reliable decisions under arbitrary distribution shift. Uncertainty assessment must therefore report both reliability and decision utility, with any formal risk-control claim supported by an explicitly defined calibration algorithm and its statistical assumptions.

These challenges motivate the investigation of complementary individual and relational evidence. Account activity rhythms describe the temporal organization of an individual account, but do not necessarily reveal whether its actions are unusually synchronized with those of other accounts. Conversely, a static interaction graph may capture relationships without representing when the associated actions occurred. Temporal co-activity modeling can represent relationships between accounts that act on shared observable objects within a temporal neighborhood. Nevertheless, temporal proximity alone is not sufficient evidence of coordinated automation: popular content, scheduled events, and legitimate communities can generate synchronized activity by chance or in response to common external stimuli. Null-model correction is therefore incorporated to estimate the extent to which observed co-activity exceeds a specified chance-based reference. Its usefulness must be established experimentally, because the validity of the correction depends

on the null model, available observations, and assumptions about the underlying activity process.

This paper investigates a campaign-shift-aware social bot detection framework that integrates account activity rhythms, temporal co-activity relationships, profile characteristics, graph-based prediction, and selective uncertainty assessment. An account-rhythm encoder summarizes individual activity patterns from timestamped events. A temporal co-activity graph represents relationships between accounts based on their actions on shared observable objects, using a multiscale temporal kernel to weight event proximity and null-model correction to discount co-activity expected under a specified reference process. A profile feature extractor represents available account attributes and missingness information. These complementary representations are integrated by a graph neural network to estimate account-level class probabilities. The resulting predictions are evaluated under random, campaign-disjoint, chronological, and cross-dataset settings, while a separate calibration and selective-prediction stage examines prediction-set reliability, automatic-decision coverage, and abstention.

The uncertainty component requires particular methodological care. We formulate campaign-unit calibration in terms of a bounded campaign-level loss and distinguish empirical reliability from formal risk control. A theoretical expected-risk guarantee is meaningful only when the implemented calibration procedure satisfies the assumptions of its associated result, including the definition of the calibration unit, the relevant exchangeability conditions, and the required properties of the loss and threshold-selection procedure. Exchangeability across calibration campaigns does not automatically imply conditional validity for each future campaign or robustness to arbitrary distribution shift. Similarly, cross-fitting or leave-campaign-out evaluation does not inherit a fixed-model conformal guarantee without an appropriate validity argument. Accordingly, the study treats measured campaign-level miscoverage and formal risk-control guarantees as separate forms of evidence.

The research is organized around eight objectives. First, it quantifies the performance difference between random account-level and campaign-disjoint evaluation to determine whether conventional splitting overestimates generalization. Second, it evaluates whether timestamp-aware coordination improves unseen-campaign detection beyond individual-feature, static-graph, and temporal-graph baselines. Third, it isolates the contribution of null-model correction through controlled comparisons of corrected and uncorrected co-activity representations; this is distinct from the second objective, which evaluates the overall contribution of temporal coordination. Fourth, it assesses campaign-unit calibration using empirical miscoverage, deviation from a prespecified target, and uncertainty across independent evaluation units, while restricting theoretical claims to justified assumptions. Fifth, it characterizes the trade-off between automatic decisions and abstention using prediction-set size, singleton rate, abstention rate, selective accuracy, and selective Macro-F1. Sixth, it tests robustness to missing events, timestamp perturbations, low-activity accounts, and synchronized legitimate communities where reliable annotations are available. Seventh, it measures graph-construction, training, and inference costs as the number of accounts and interactions increases. Eighth, it evaluates cross-dataset transfer and the associated changes in detection, calibration, and selective-prediction behavior.

The principal contributions are summarized as follows:

- **Campaign-aware evaluation:** We establish a leakage-controlled evaluation protocol that distinguishes random account-level performance from campaign-disjoint, chronological, and cross-dataset generalization, thereby enabling a more systematic assessment of performance under different forms of distributional variation.

- **Temporal coordination representation:** We develop a representation that integrates individual activity rhythms, timestamp-aware co-activity relationships, null-model correction, and account-profile characteristics. Controlled comparisons and component ablations are designed to quantify the incremental contribution of each component.
- **Campaign-unit uncertainty assessment:** We formulate campaign-level miscoverage using a bounded loss and specify the assumptions relevant to campaign-unit calibration. The evaluation distinguishes empirical reliability from formal risk-control guarantees rather than inferring theoretical validity from observed coverage alone.
- **Selective prediction under campaign shift:** We integrate selective prediction into the evaluation framework to quantify the relationship between prediction-set reliability, automatic-decision coverage, and abstention, including the implications for human review.
- **Systematic experimental assessment:** We define experiments covering campaign-shift degradation, coordination modeling, null-correction ablations, calibration, robustness, computational scalability, and cross-dataset generalization, with statistical analysis conducted at the appropriate evaluation-unit level.

The contribution of this work is not based merely on combining graph neural networks, temporal features, and conformal prediction, since these methodological families have established precedents. Instead, the study examines their interaction within a campaign-aware evaluation framework and tests three distinct questions: whether temporal coordination improves detection on unseen campaigns, whether the improvement persists under controlled component comparisons, and whether selective uncertainty assessment provides useful reliability–coverage trade-offs under explicitly stated assumptions. The strength of these contributions must be established through comparison with the closest existing methods, verified data, controlled experiments, and statistical evidence. No numerical improvement or formal guarantee is inferred solely from the proposed architecture.

The remainder of this paper is organized as follows. Section 2 reviews related work and identifies the research gap. Section 3 summarizes the system architecture, and Section 4 defines the problem and its assumptions. Section 5 describes the proposed method. Section 6 presents the datasets, evaluation protocols, baselines, metrics, and ablation studies. Section 7 summarizes the framework and identifies directions for future work.

2 Related Work and Research Gap

Social bot detection has evolved from account-level classification to multimodal representation learning, heterogeneous graph analysis, dynamic network modeling, and coordinated group detection. Despite these advances, predictive performance on conventional benchmark partitions does not necessarily establish generalization to previously unseen campaigns or reliable automated decisions. This section reviews representative studies, classifies their methodological contributions, and positions the proposed campaign-shift-aware framework relative to existing approaches.

Individual-feature and multimodal detection. Cresci [1] reviews a decade of social bot detection research and discusses the evolution of detection strategies in response to increasingly sophisticated automated accounts. Feng et al. [2] introduce TwiBot-20, a benchmark containing account profiles, tweets, and network relationships. Their evaluation demonstrates the importance of diverse and sufficiently informative benchmark data for reproducible model comparison. Feng

Table 1: Classification of representative social bot detection studies according to behavioral representation and evaluation capability.

Behavioral representation	**Detection and representation methods**	**Generalization and reliability considerations**
Individual and multimodal features	Cresci [1]; TwiBot-20 [2]; SATAR [3]	Feature robustness, account diversity, and evaluation beyond conventional account-level partitions.
Temporal behavior and coordination	Chavoshi et al. [4]; Khaund et al. [5]	Distinguishing recurring or coordinated activity from generalization to previously unseen campaigns.
Heterogeneous graph learning	BotRGCN [6]; RGT [7]; TwiBot-22 [8]	Relational representation quality, benchmark reproducibility, and campaign-disjoint evaluation.
Dynamic graph and group interactions	BotDGT [9]; BotSTIP [10]	Temporal network evolution, group interactions, and generalization beyond the original evaluation setting.
Conformal prediction and risk control	Angelopoulos and Bates [11]; Angelopoulos et al. [12]	Calibration assumptions, bounded-loss guarantees, prediction-set coverage, and selective utility.
Proposed framework	**Temporal co-activity, null-model correction, profile features, and relational prediction**	**Campaign-disjoint evaluation, campaign-unit calibration, and selective prediction under explicit assumptions.**

et al. [3] propose SATAR, a self-supervised account representation learning approach that uses multiple account information sources to improve bot detection. These studies establish the value of multimodal features and learned account representations. However, the use of diverse features alone does not establish generalization to campaigns excluded from training. In comparison, the proposed framework investigates whether explicit temporal coordination provides additional information beyond individual account representations under campaign-disjoint evaluation.

Temporal behavior and coordinated activity. Chavoshi et al. [4] develop a system for detecting and archiving bots using temporal activity patterns and correlations between accounts. Their work illustrates how synchronized behavior can provide evidence that is difficult to obtain from isolated account characteristics. Khaund et al. [5] survey social bot coordination during online campaigns and describe methods for analyzing coordinated account behavior. These studies establish that temporal synchronization and coordinated activity are existing research directions. Consequently, the novelty of the proposed framework cannot rest on introducing coordination analysis alone. Instead, it must demonstrate whether timestamp-aware co-activity construction and null-model correction improve detection on campaigns excluded from model development.

Heterogeneous graph representation learning. Feng et al. [6] introduce BotRGCN, which combines relational graph convolution with heterogeneous social relationships and multimodal account information. Feng et al. [7] develop relational graph transformer methods that exploit different types of social relations for bot detection. Feng et al. [8] introduce TwiBot-22, a large-

scale graph-based benchmark that supports the systematic comparison of detection methods across multiple datasets. Collectively, these contributions demonstrate the importance of relational information, heterogeneous graph representations, and standardized evaluation. Nevertheless, a graph-based architecture does not automatically establish robustness to campaign shift. The proposed framework therefore requires controlled comparisons against heterogeneous graph baselines to isolate the additional value of timestamp-aware co-activity modeling.

Dynamic graph and group-interaction modeling. He et al. [9] propose BotDGT, which models the evolving structure of social networks through dynamic graph transformers. The method integrates structural information from historical graph snapshots with temporal representations of network evolution. Liu and Ma [10] introduce BotSTIP, which uses spatiotemporal group interactions, hypergraph-based representations, and temporal attention to model relationships among social accounts. BotSTIP is particularly relevant because it already addresses temporal and group-level interactions. Accordingly, the proposed framework must be compared directly with such methods rather than claiming novelty from temporal and relational feature integration alone. Its potential distinction lies in the combination of null-corrected temporal co-activity, explicit campaign-disjoint evaluation, and campaign-unit uncertainty assessment. Whether this combination provides an empirical advantage remains to be established experimentally.

Conformal prediction and uncertainty control. Angelopoulos and Bates [11] provide a comprehensive introduction to conformal prediction and explain the assumptions under which prediction sets achieve marginal coverage guarantees. Angelopoulos et al. [12] extend conformal calibration to suitable bounded loss functions through conformal risk control. These results provide a foundation for evaluating prediction reliability beyond classification accuracy. However, marginal coverage and expected-loss control are distinct properties, and neither automatically implies conditional reliability for every previously unseen campaign. The proposed framework therefore defines the calibration unit, bounded loss, and exchangeability assumptions explicitly. Campaign-level risk control must be justified for the specified calibration procedure, while account-level coverage, selective accuracy, and abstention are evaluated separately.

Research gap and positioning. The literature establishes that multimodal features, temporal behavior, heterogeneous graphs, dynamic network models, and conformal prediction are valuable methodological components. However, these components address different aspects of detection and cannot be assumed to provide campaign-level reliability when combined without further analysis. Three issues motivate the proposed investigation: the need to distinguish random account-level performance from campaign-disjoint generalization; the need to measure the incremental value of temporal coordination against existing dynamic and group-aware models; and the need to evaluate selective prediction under calibration assumptions consistent with the claimed risk guarantee.

The proposed framework investigates these issues through a unified experimental design comprising temporal co-activity representation, relational prediction, campaign-disjoint and chronological evaluation, campaign-unit calibration, and selective decision-making. Its contribution is not the isolated introduction of temporal graphs or conformal prediction, but the explicit formulation and evaluation of their roles in campaign-shift-aware social bot detection. The novelty and practical value of this formulation must be demonstrated through verified comparisons, leakage-controlled experiments, component ablations, and statistical uncertainty analysis.

3 System Architecture

The proposed system architecture integrates temporal behavior analysis, coordination-aware graph construction, relational learning, campaign-aware evaluation, and selective prediction to detect social bots under campaign shift. Its purpose is to determine whether temporal coordination improves detection on previously unseen campaigns and whether uncertainty-aware prediction can identify accounts suitable for automatic classification. The overall processing pipeline is illustrated in Fig. 1.

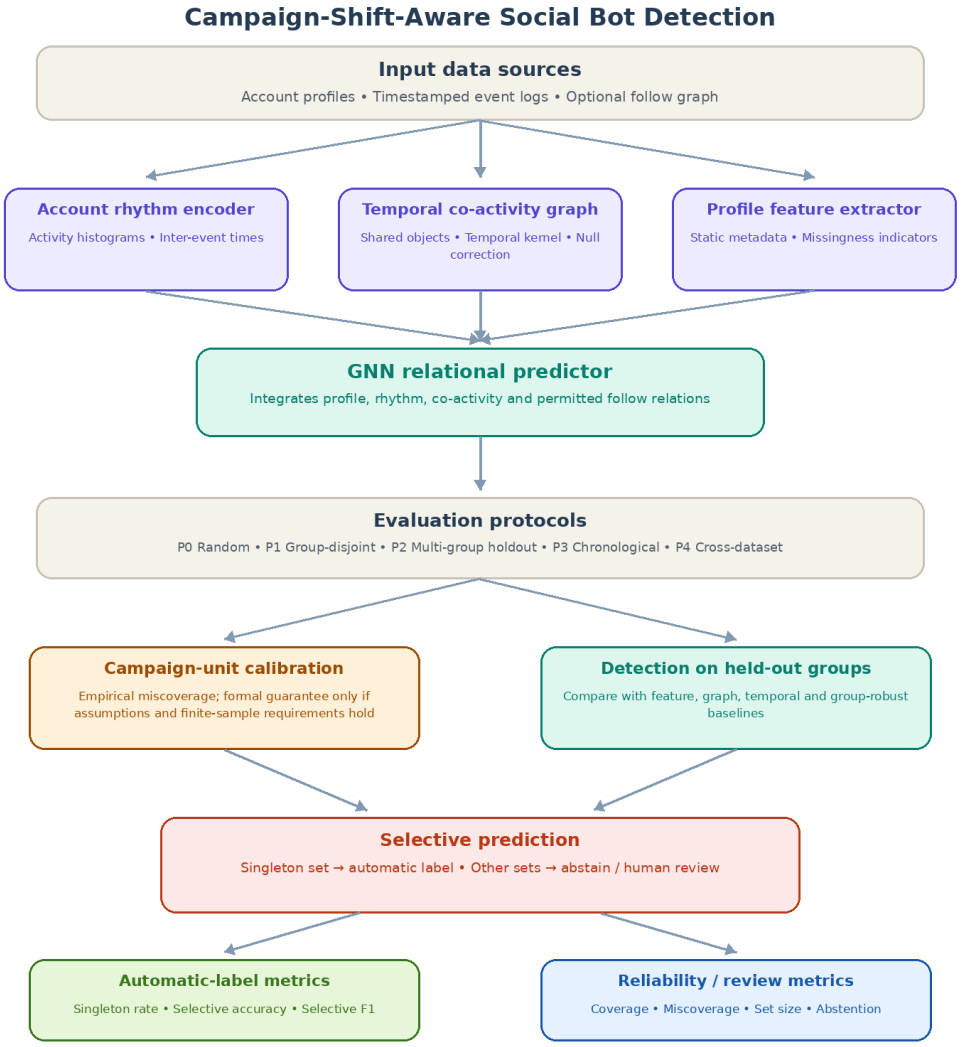


Figure 1: Overview of the campaign-shift-aware social bot detection framework. Calibration and empirical evaluation are separate stages; any risk guarantee depends on the assumptions of the selected calibration procedure.

Input Data Sources: The purpose of this component is to provide the information required for bot detection. It collects account profiles, timestamped activity events, optional follow relationships, ground-truth labels, and campaign identifiers. These data provide static account characteristics, temporal behavior, network relationships, and supervision for model training. Campaign identifiers also enable the evaluation of generalization to previously unseen campaigns.

Account Rhythm Encoder: This component identifies temporal regularities in individual

account activity. It processes timestamped events to generate weekly activity histograms, inter-event-time distributions, and spectral summaries when sufficient observations are available. These features represent how accounts distribute their activity over time and provide behavioral evidence that complements static profile information. Missingness indicators distinguish insufficient observations from genuine zero-valued measurements.

Co-Activity Graph Builder: This component identifies potential coordination between accounts interacting with shared objects at similar times. It computes temporal proximity using a multiscale kernel and constructs weighted relationships between accounts. A time-permutation null model reduces associations attributable to chance temporal overlap under the specified null assumptions. The resulting graph provides relational evidence of unusual co-activity that cannot be captured by analyzing accounts independently.

Profile Feature Extractor: This component represents the static characteristics of each account. It processes available metadata, such as account age, follower and following counts, and profile completeness, together with missingness indicators. The resulting representation provides complementary evidence about account characteristics and helps the detector combine profile-based and temporal behavioral information.

GNN Relational Predictor: This component combines individual account characteristics with information from related accounts to classify human and bot accounts. It receives profile features, temporal representations, coordination statistics, and the constructed graph. A graph neural network performs message passing over the available co-activity and follow relationships to learn relational account representations. A classification layer converts these representations into class probabilities, enabling the detector to exploit coordination patterns that may be overlooked by feature-only classifiers.

Evaluation Protocol: This component determines whether the detector generalizes beyond the accounts and campaigns observed during training. It establishes random account-level, campaign-disjoint, multi-campaign holdout, chronological, and cross-dataset evaluation settings, where applicable. These settings measure different forms of generalization and help reveal performance inflation caused by campaign-specific information leakage. Training, calibration, and final test data are separated to ensure that test labels do not influence model fitting or model selection.

Campaign-Unit Calibration: This component selects a decision threshold using held-out calibration units, such as campaigns, without using final test labels. It evaluates the prescribed bounded loss across candidate thresholds and applies the selected conformal risk-control procedure. Its purpose is to provide a statistically principled basis for uncertainty-aware decisions under the assumptions of the corresponding method. If the target risk is infeasible or the required exchangeability assumptions are not justified, this limitation is explicitly reported.

Detection on Test Campaigns: This component evaluates the trained detector on previously unseen test accounts. It receives the fixed trained model and the permitted test features and graph information, then produces human and bot class probabilities. The predictions are compared with suitable baseline methods using Macro-F1, precision–recall area under the curve (PR-AUC), and Matthews correlation coefficient (MCC). This component quantifies detection performance under each evaluation protocol without using test labels to adjust the model or calibration threshold.

Selective Prediction Output: This component determines whether each account can be classified automatically or requires further review. It receives the prediction set generated by the calibrated decision rule. If the set contains exactly one class, the corresponding label is emitted

automatically. If the set contains multiple classes, the system abstains and refers the account for human review. This mechanism makes the trade-off between automatic classification and uncertainty explicit.

Output Metrics: This component measures both classification performance and the reliability of selective decisions. It uses ground-truth test labels, predictions, prediction sets, and abstention decisions to calculate Macro-F1, PR-AUC, MCC, singleton rate, selective accuracy, abstention rate, prediction-set size, empirical coverage, and miscoverage. Reporting these measures separately distinguishes overall discrimination performance from the reliability and coverage of automatic decisions.

The interaction among these components forms an end-to-end pipeline for campaign-aware social bot detection. The account rhythm encoder and profile feature extractor represent individual behavior, while the co-activity graph builder captures temporal relationships. The GNN relational predictor integrates these representations to generate class probabilities. The evaluation protocol establishes the generalization setting, and the calibration and selective prediction components determine which predictions can be issued automatically. Algorithm 1 in Section 5 summarizes the complete processing sequence, including data preparation, model training, calibration, test-time prediction, and performance evaluation.

The detailed processing sequence is specified once in Algorithm 1 in Section 5; this architecture section is intended as a high-level overview rather than a second algorithmic specification.

The complete execution sequence is specified once in Algorithm 1 in Section 5. This section provides the conceptual architecture; the detailed algorithm is not duplicated.

4 Problem Formulation

4.1 Problem Definition and Data Representation

Let $\mathcal{U}$ denote a collection of observed social media accounts. Each account $u \in \mathcal{U}$ is represented by its profile attributes, temporal activity, and, where available, relational information. Its observable event sequence is defined as

$$\mathcal{E}_u = \{(t_k, a_k, o_k)\}_{k=1}^{n_u}, \tag{1}$$

where t_k denotes the timestamp of event k, a_k represents the event or action type, o_k identifies the associated object, and n_u is the number of observed events. Depending on the dataset, an object may correspond to a post, hashtag, or target account. The profile representation is denoted by $\mathbf{p}_u \in \mathbb{R}^{d_p}$, where d_p is the number of available profile features. The binary class label is $y_u \in \{0, 1\}$, with 0 representing a human account and 1 representing a social bot.

When follow relationships are available, they are represented by $\mathcal{G}_f = (\mathcal{U}, \mathcal{E}_f)$, where $\mathcal{E}_f$ denotes the observed follow edges. Follow relationships provide relational context but are not treated as evidence of temporal interaction unless corresponding timestamps are available. All representations are constructed exclusively from fields supported by the selected dataset.

Let $c(u)$ denote the independently annotated campaign membership of account u, whenever reliable campaign annotations are available. The set of accounts belonging to campaign g is denoted by $\mathcal{U}_g$. Campaign membership is used to define evaluation groups and calibration units; it is not assumed to be available as a predictive feature at inference time. If campaigns are approximated using collection batches or behavioral heuristics, these groups must be identified as proxies rather than verified campaigns.

The central problem is that a detector evaluated using randomly partitioned accounts may exploit campaign-specific regularities shared across training and test data. Consequently, its measured performance may not represent its ability to identify bots in previously unseen campaigns. The proposed study therefore examines whether temporal coordination modeling improves detection under campaign-disjoint evaluation and whether uncertainty-aware prediction can reduce unreliable automatic decisions.

4.2 Campaign-Shift-Aware Generalization

Let $\mathcal{C}_{\text{tr}}$ and $\mathcal{C}_{\text{te}}$ denote the sets of campaigns represented in the training and test partitions, respectively. The campaign-disjoint evaluation protocol imposes

$$\mathcal{C}_{\text{tr}} \cap \mathcal{C}_{\text{te}} = \varnothing. \tag{2}$$

This constraint prevents annotated campaigns from appearing in both partitions. It must be complemented by duplicate removal, identifier-overlap checks, and feature-construction rules that prevent information from the test period or test labels from entering training.

Let $\mathcal{D}_{\text{tr}}$ denote the training distribution and $\mathcal{D}_{\text{te}}$ the test distribution. Campaign shift refers to a change in the distribution of account features, coordination patterns, or class-conditional behavior between training and previously unseen campaigns. In general,

$$\mathcal{D}_{\text{tr}} \neq \mathcal{D}_{\text{te}}, \tag{3}$$

although the precise form and magnitude of this difference must be established empirically. This notation expresses the motivation for campaign-shift-aware evaluation; it does not imply that all distributional changes can be detected or corrected.

Three complementary evaluation settings are considered. First, random account splitting measures performance when accounts may share campaign-specific characteristics across partitions. Second, campaign-disjoint splitting evaluates generalization to unseen campaigns. Third, chronological evaluation restricts feature construction to information available at the prediction time. Cross-dataset evaluation provides an additional test of transfer to a separately collected corpus. These settings are analyzed separately because they represent different sources of distributional variation and are not interchangeable.

4.3 Account Representation and Detection

For each account u, the proposed detector constructs a representation using three complementary evidence sources: profile characteristics, temporal activity patterns, and temporal coordination with other accounts. Let $\mathbf{h}_u^p$, $\mathbf{h}_u^r$, and $\mathbf{h}_u^c$ denote the corresponding profile, rhythm, and coordination representations. The integrated account representation is defined as

$$\mathbf{x}_u = \phi_\psi \left(\mathbf{h}_u^p \| \mathbf{h}_u^r \| \mathbf{h}_u^c \right), \tag{4}$$

where $\|$ denotes feature concatenation and ϕ_ψ denotes the feature-integration function, parameterized separately from the classifier. The representation is subsequently supplied to a classifier parameterized by θ (with classifier parameters distinct from ψ):

$$f_\theta(\mathbf{x}_u) = \widehat{\mathbf{p}}_u = (\widehat{p}_{u,0}, \widehat{p}_{u,1}), \qquad \sum_{y \in \{0,1\}} \widehat{p}_{u,y} = 1, \tag{5}$$

where $\widehat{p}_{u,y}$ is the estimated probability that account u belongs to class y.

The conventional class prediction is

$$\widehat{y}_u = \arg\max_{y \in \{0,1\}} \widehat{p}_{u,y}. \tag{6}$$

However, the most probable class is not necessarily sufficiently reliable for automatic deployment. The detector therefore also supports uncertainty-aware prediction, in which a prediction set is constructed and accounts with unresolved predictions may be deferred for human review.

The temporal coordination representation is intended to capture synchronized or closely related activity that cannot be fully described by an account's individual activity rhythm. Its effectiveness is not assumed in advance. It is evaluated through comparisons with profile-only, rhythm-only, and coordination ablations under identical data partitions and evaluation conditions. This design directly tests whether coordination information contributes additional predictive value, particularly when the test campaigns are absent from training.

4.4 Prediction Sets and Account-Level Reliability

Let $\Gamma_\tau(\mathbf{x}_u) \subseteq \{0, 1\}$ denote the prediction set returned for account u, where τ is a threshold selected by the specified set-construction and calibration procedure. A singleton set supports an automatic class assignment, whereas a set containing both classes indicates that the system abstains from making an automatic decision. An empty set, if permitted by the chosen construction, also requires review.

The account-level miscoverage indicator is defined as

$$\ell_u(\tau) = \mathbb{I}\left[y_u \notin \Gamma_\tau(\mathbf{x}_u)\right], \tag{7}$$

where $\mathbb{I}[\cdot]$ is the indicator function. If the prediction sets are nested and become smaller as τ increases, then $\ell_u(\tau)$ is non-decreasing in τ. This monotonicity is conditional on the actual set-construction rule and must be verified for its implementation.

For a campaign g containing $|\mathcal{U}_g|$ evaluated accounts, campaign-level miscoverage is defined as

$$L_g(\tau) = \frac{1}{|\mathcal{U}_g|} \sum_{u \in \mathcal{U}_g} \ell_u(\tau), \qquad 0 \le L_g(\tau) \le 1. \tag{8}$$

This bounded quantity measures the fraction of accounts in a campaign whose true labels are excluded from their prediction sets. Averaging within each campaign gives every campaign a well-defined risk value regardless of its number of accounts. However, the definition alone does not establish statistical exchangeability or a formal risk guarantee.

Let $G_1, \ldots, G_m$ denote the calibration campaigns and G_{m+1} a future campaign. A campaign-unit calibration procedure uses the calibration data to select a threshold $\widehat{\tau}$. For a prespecified risk tolerance $\alpha \in (0, 1)$, the desired expected campaign-level risk target is

$$\mathbb{E}\left[L_{G_{m+1}}(\widehat{\tau})\right] \le \alpha. \tag{9}$$

The expectation is over the relevant calibration and future campaign sampling process. This inequality is a target for evaluation, not an established guarantee of the proposed detector. Its validity depends on the exact calibration algorithm, the definition of the calibration unit, and the assumptions required by the corresponding statistical result. In particular, the necessary

exchangeability conditions must be justified rather than inferred from campaign-disjoint partitioning alone. Arbitrary campaign shift does not automatically satisfy these conditions, and the target does not imply a guarantee for every individual unseen campaign.

Accordingly, campaign-level miscoverage will be reported empirically on held-out campaigns. A formal risk-control claim will be made only if the implemented calibration procedure and its assumptions support the stated result. Otherwise, the analysis will be interpreted as an empirical assessment of campaign-level reliability.

4.5 Selective Prediction and Review Burden

For operational use, an account is eligible for automatic labeling when its prediction set contains exactly one class. Define the automatic-label indicator as

$$A_u = \mathbb{I}\left[|\Gamma_\tau(\mathbf{x}_u)| = 1\right]. \tag{10}$$

Accounts whose prediction sets are not singletons are deferred for review. For a test set containing N accounts, the singleton rate and abstention rate are

$$\mathrm{SR} = \frac{1}{N}\sum_{u=1}^{N} A_u, \qquad \mathrm{AR} = 1 - \mathrm{SR}. \tag{11}$$

These measures quantify the fraction of accounts receiving automatic labels and the corresponding review burden.

The correctness of automatic decisions is measured separately using selective accuracy:

$$\mathrm{SA} = \frac{\sum_{u=1}^{N} A_u\, \mathbb{I}[\widehat{y}_u = y_u]}{\sum_{u=1}^{N} A_u}, \tag{12}$$

provided that at least one account receives an automatic label. For singleton prediction sets, $\widehat{y}_u$ is the unique class in $\Gamma_\tau(\mathbf{x}_u)$. If the classifier's highest-probability prediction is not the unique class returned by the prediction-set procedure, the automatic label must instead be defined consistently as that unique class.

Empirical prediction-set coverage is defined as

$$\mathrm{Coverage} = \frac{1}{N}\sum_{u=1}^{N} \mathbb{I}\left[y_u \in \Gamma_\tau(\mathbf{x}_u)\right]. \tag{13}$$

Coverage measures whether the true class is included in the prediction set, whereas selective accuracy measures the correctness of decisions among automatically labeled accounts. A non-singleton set can contain the true class without supporting an automatic decision. Therefore, coverage, campaign-level miscoverage, selective accuracy, singleton rate, and abstention are distinct measures and must not be interpreted as substitutes for one another.

4.6 Research Objectives and Evaluation Criteria

The formulation establishes eight linked evaluation objectives.

O1: Quantify campaign-shift degradation. Measure the change in detection performance between random account splits and campaign-disjoint splits using Macro-F1, PR-AUC, and complementary classification metrics.

O2: Evaluate unseen-campaign detection. Determine whether the proposed representation maintains discriminative performance on campaigns excluded from training.

O3: Isolate the contribution of temporal coordination. Compare the complete detector with profile-only, rhythm-only, and coordination-removal variants to quantify the incremental contribution of temporal coordination modeling.

O4: Assess prediction reliability. Measure empirical prediction-set coverage and campaign-level miscoverage on held-out campaigns, distinguishing observed performance from any formally justified risk-control guarantee.

O5: Quantify selective-prediction utility. Analyze the trade-off between selective accuracy, singleton rate, and abstention as the prediction-set threshold varies.

O6: Evaluate robustness. Assess sensitivity to missing events, timestamp perturbations, low-activity accounts, and legitimate synchronized behavior.

O7: Measure computational cost. Report training and inference time, memory consumption, and the computational overhead introduced by temporal coordination modeling and calibration.

O8: Examine cross-dataset generalization. Evaluate transfer to an independently collected dataset under a documented label-mapping and feature-compatibility protocol.

The evaluation is multi-objective rather than a claim that one scalar score captures all deployment requirements. Computational cost is reported as a tuple

$$\mathbf{C}_{\text{comp}} = (T_{\text{graph}}, T_{\text{train}}, T_{\text{infer}}, M_{\text{peak}}), \tag{14}$$

where the terms denote graph-construction time, model-training time, inference time per account (or batch, stated explicitly), and peak memory consumption. These quantities are reported separately with the hardware and workload specified; they are not combined using arbitrary weights. Detection quality, prediction-set reliability, selective utility, and computational cost are therefore interpreted jointly, because optimizing one criterion need not improve the others.

The experiments are designed to determine whether temporal coordination modeling improves detection on unseen campaigns, whether the resulting predictions remain empirically reliable, and whether selective prediction offers a useful balance between automatic decision quality and review workload. Controlled ablations, leakage-aware data partitions, and separately reported computational measurements provide the evidence needed to assess these objectives.

5 Proposed System

The proposed Campaign-Shift-Aware Social Bot Detection framework is designed to identify social bots while evaluating the reliability of detection on previously unseen campaigns. Conventional account-level classification can achieve high performance under random data splits when training and test accounts share campaign-specific behavioral patterns. However, such performance does not necessarily demonstrate generalization to campaigns with different activity rhythms, coordination structures, or account characteristics. The proposed system addresses this limitation by combining individual account representations, temporal coordination modeling, graph-based relational prediction, campaign-aware evaluation, and selective prediction.

As established in the problem formulation, each account $u \in \mathcal{U}$ is represented by a profile vector $\mathbf{p}_u$, a timestamped event sequence $\mathcal{E}_u$, and a binary label $y_u \in \{0, 1\}$, where the two classes denote human and bot accounts. Campaign membership is represented by $c(u)$ whenever

this information is available. The objective is to learn a predictor that exploits both individual behavioral evidence and relationships among accounts while evaluating its performance under explicitly defined campaign and temporal partitions. In addition, the system must distinguish accounts that can be automatically labeled from those for which the available evidence does not support a unique decision.

The proposed system comprises account-level representation learning, temporal co-activity graph construction, null-model correction, relational classification, campaign-aware evaluation, calibration, and selective prediction. The account-level modules characterize individual behavior, whereas the graph module captures temporal relationships between accounts. The relational predictor integrates these sources of evidence, and the final decision modules assess generalization and determine whether a prediction should be issued automatically. The complete execution procedure is summarized in Algorithm 1.

Input representation and campaign-aware partitioning: The input component receives account profiles, timestamped events, optional follow relationships, ground-truth labels, and campaign identifiers. Each event is represented as (t_k, a_k, o_k), where t_k denotes the event timestamp, a_k denotes the action type, and o_k denotes the associated observable object. Depending on dataset availability, an object may represent a URL, hashtag, or another shared resource. The optional follow graph is denoted by $\mathcal{G}_f = (\mathcal{U}, \mathcal{E}_f)$.

Before feature construction, the data are partitioned according to the intended evaluation setting. For campaign-disjoint evaluation, training and test campaign sets must satisfy $\mathcal{C}_{\text{tr}} \cap \mathcal{C}_{\text{te}} = \varnothing$. For chronological evaluation, only events available by the prediction cutoff are used to generate features and graph edges. Training, calibration, and final test data are kept separate according to the selected evaluation protocol. The predictor is trained using labeled training accounts, calibration labels are reserved for the prescribed threshold-selection procedure, and test labels are withheld until final evaluation. This separation directly operationalizes the campaign-disjoint and information-leakage constraints specified in the problem formulation.

Account rhythm representation: The account rhythm encoder characterizes the temporal organization of individual account activity. Its input is the event sequence $\mathcal{E}_u$ of each account. It extracts complementary statistics that describe activity distribution and temporal regularity. Weekly activity histograms represent how activity is distributed across the week, while inter-event-time distributions characterize the intervals between consecutive events. Spectral summaries can additionally capture periodic patterns when the event history is sufficiently long and informative.

Let $\mathbf{r}_u$ denote the resulting temporal representation. The construction can be expressed as

$$\mathbf{r}_u = \phi_{\text{rhythm}}(\mathcal{E}_u; T), \tag{15}$$

where ϕ_{rhythm} denotes the specified feature-extraction procedure and T denotes the prediction cutoff. The cutoff ensures that no event occurring after the permitted observation time contributes to the representation. Missingness indicators distinguish unavailable or insufficient temporal evidence from genuine zero-valued statistics. Any normalization parameters are estimated from training data and subsequently applied unchanged to calibration and test accounts.

This representation provides individual behavioral evidence but does not, by itself, establish whether an account participates in coordinated activity. The next component therefore models temporal relationships among accounts.

Temporal co-activity graph construction: The co-activity graph builder identifies pairs of accounts that interact with shared objects within temporally proximate intervals. For account

u, let $\mathcal{E}_u(o)$ denote the timestamps of events associated with object o. A multiscale kernel measures temporal proximity at different time scales:

$$\kappa_{\boldsymbol{\theta}}(\delta) = \sum_{m=1}^{M} \pi_m \exp\left(-\frac{\delta}{\lambda_m}\right), \tag{16}$$

subject to $\pi_m \geq 0$, $\sum_{m=1}^{M} \pi_m = 1$, and $\lambda_m > 0$. Here, δ is the absolute time difference between two events, π_m denotes the weight assigned to temporal scale m, and λ_m controls the decay rate at that scale. Shorter decay parameters emphasize near-simultaneous activity, whereas longer parameters allow associations across wider intervals. The weights and decay parameters determine how evidence from different temporal scales is combined. In this implementation, these parameters are selected using training/validation data and then fixed before final testing; the discrete null estimation and graph sparsification steps are not treated as end-to-end differentiable operations.

The raw co-activity score between accounts u and v is calculated by aggregating temporal proximity over their shared observable objects:

$$\widetilde{w}_{uv} = \sum_{o \in \mathcal{O}_{uv}} \sum_{t \in \mathcal{E}_u(o)} \sum_{t' \in \mathcal{E}_v(o)} \kappa_{\boldsymbol{\theta}}(|t - t'|), \tag{17}$$

where $\mathcal{O}_{uv}$ denotes the set of shared objects observed for the two accounts within the permitted observation window. The score increases when the accounts repeatedly interact with shared objects at temporally proximate times. This operation constructs a candidate relationship between accounts rather than directly assigning a bot label.

The kernel in Equation 16 is a temporal proximity function. It is not a Hawkes-process model and does not establish causal excitation between account activities. Its purpose is to represent multiscale synchronization in a form that can be incorporated into a relational classifier.

Null-model correction and graph sparsification: Raw co-activity can be inflated by highly popular objects, common events, or chance temporal overlap. To reduce this effect, the system estimates a null distribution using constrained time randomization within predefined observation intervals. Where the dataset supports it, randomization is stratified by object and coarse time window so that broad object popularity and time-varying activity are not indiscriminately destroyed. The exact randomization scheme is documented and validated against synchronized legitimate activity where suitable annotations are available. The resulting null distribution represents co-activity expected under that specific procedure, not a universal model of organic behavior; a simple within-account permutation alone cannot fully control for shared external events.

Let μ^0_{uv} and σ^0_{uv} denote the estimated mean and standard deviation of the null scores for account pair (u, v). The null-corrected edge weight is defined as

$$w_{uv} = \max\left(0, \frac{\widetilde{w}_{uv} - \mu^0_{uv}}{\sigma^0_{uv} + \varepsilon}\right), \tag{18}$$

where $\varepsilon > 0$ prevents numerical instability when the estimated null standard deviation is small. Pairs with insufficient null variation or too few shared events are handled using a prespecified minimum-support rule and are not allowed to acquire arbitrarily large weights solely because σ^0_{uv} is near zero. The corrected score measures the excess of observed co-activity relative to the chosen null distribution. A positive corrected score indicates that the observed value exceeds the

estimated null mean; it does not, on its own, establish statistical significance or malicious intent.

The graph is sparsified by removing edges below a predetermined threshold and retaining at most k neighbors per account. These parameters, the number of permutations, random seeds, active-interval definition, and object-frequency cap are documented and selected using training data only. Where computational approximations are used for popular objects, their effect is validated against more direct permutation estimates on training data. The resulting weighted graph is denoted by $\mathcal{G}_c = (\mathcal{U}, \mathcal{E}_c, \mathbf{W}_c)$, where $\mathcal{E}_c$ contains the retained edges and $\mathbf{W}_c$ contains their corrected weights.

This procedure operationalizes the coordination component of the problem formulation. It reduces the influence of some forms of chance co-activity while preserving relationships that are unusual under the selected null model. Its effectiveness must be established empirically through comparisons with uncorrected and fixed-window graph constructions.

Coordination statistics and feature integration: The graph structure provides relational evidence, while account-level coordination statistics summarize the local structure around each account. The system derives measures such as weighted degree, local clustering, and synchronized-event fraction. Let $\mathbf{q}_u$ denote the vector of these coordination statistics. The weighted degree summarizes the strength of an account's retained temporal relationships, local clustering measures connectivity among its neighbors, and synchronized-event fraction characterizes the proportion of its observed activity associated with temporally coordinated events under the chosen definition.

The initial representation combines profile features, temporal features, and coordination statistics:

$$\mathbf{h}_u^{(0)} = \phi_{\text{proj}}\left(\mathbf{p}_u \| \mathbf{r}_u \| \mathbf{q}_u\right), \tag{19}$$

where $\|$ denotes concatenation and ϕ_{proj} denotes a learned projection into a common representation space. The profile vector is processed using the training-fitted transformations, while missingness indicators are retained where necessary. The projection allows heterogeneous features to be integrated into a common input for relational learning. Its architecture and dimensions are specified in the experimental configuration.

This integration is important because static account characteristics, individual activity rhythms, and temporal coordination describe different aspects of behavior. Combining them allows the classifier to use multiple sources of evidence without assuming that any one feature group is sufficient.

Graph neural network relational prediction: The relational predictor uses the initial account representations and graph relationships to learn account embeddings for binary classification. Its purpose is to combine individual behavioral evidence with information from connected accounts. If the follow graph is available and permitted by the dataset, it is incorporated as a separate relation type rather than being treated as equivalent to temporal coordination.

Let $\mathcal{R}$ denote the set of available graph relation types and $\mathcal{N}_r(u)$ the neighbors of account u under relation r. At message-passing layer ℓ, the representation is updated as

$$\mathbf{h}_u^{(\ell+1)} = \sigma\left(\mathbf{W}_s^{(\ell)}\mathbf{h}_u^{(\ell)} + \sum_{r\in\mathcal{R}}\sum_{v\in\mathcal{N}_r(u)} \alpha_{uv}^{(r)}\mathbf{W}_r^{(\ell)}\mathbf{h}_v^{(\ell)}\right), \tag{20}$$

where $\mathbf{W}_s^{(\ell)}$ transforms the account's own representation, $\mathbf{W}_r^{(\ell)}$ transforms neighbor representations for relation r, $\alpha_{uv}^{(r)}$ is the normalized contribution of neighbor v, and $\sigma(\cdot)$ is a nonlinear activation function. The exact neighbor-weighting and normalization rule is fixed in

the implementation and documented in the experimental configuration.

After the final message-passing layer L, a classification head produces the human and bot probabilities:

$$\widehat{\mathbf{p}}_u = \mathrm{softmax}\left(\mathbf{W}_{\mathrm{cls}}\mathbf{h}_u^{(L)} + \mathbf{b}_{\mathrm{cls}}\right), \tag{21}$$

where $\mathbf{W}_{\mathrm{cls}}$ and $\mathbf{b}_{\mathrm{cls}}$ are the classifier parameters. The predicted class is the class with the largest estimated probability when a conventional hard prediction is required.

The network is trained using labeled training accounts and a specified classification loss. Model selection is performed without consulting final test labels, and the trained parameters are fixed before calibration. To isolate the contribution of relational modeling, the proposed predictor is compared with a feature-only classifier and a fixed-window graph baseline. Additional ablations remove null correction, temporal rhythm features, coordination statistics, or optional follow relationships. These comparisons determine whether improvements are attributable to the proposed temporal coordination representation rather than simply to the use of a larger model.

Campaign-aware evaluation and leakage control: The evaluation component operationalizes the generalization conditions defined in the problem formulation. Random account-level evaluation provides a reference condition, while campaign-disjoint evaluation tests whether the model generalizes to campaigns absent from training. Chronological evaluation restricts feature and edge construction to observations available before the prediction cutoff. Cross-dataset evaluation assesses transfer where compatible labels, attributes, and observation protocols are available.

These settings are reported separately because they measure different forms of generalization. In transductive network evaluation, observable test-account relationships may be used only when they are available under the declared deployment protocol; test labels remain unavailable. In isolated-account evaluation, test-neighbor information is excluded and predictions rely on account-level features. Shared objects across partitions are not automatically leakage, but all cross-partition relationships must follow the declared protocol and be audited. For the primary campaign-disjoint inductive analysis, graph construction and message passing are restricted to the permitted training, calibration, or test partition; cross-partition edges are excluded. Any separate transductive analysis is labeled explicitly and is not used to support an independence-based calibration guarantee unless the required assumptions are established. Preprocessing, hyperparameter selection, graph-construction settings, and model selection are completed without using final test labels.

Campaign-unit risk calibration: The calibration component addresses the uncertainty-control objective by selecting a decision parameter using a held-out calibration partition and a prespecified bounded loss. It operates on predictions from the fixed trained model; calibration accounts are not used to update the predictor. When campaigns are selected as calibration units, the method requires an explicit justification for treating those units as exchangeable under the chosen risk-control procedure. Campaign membership alone is insufficient to establish exchangeability.

Let $\Gamma_q(\mathbf{x}_u)$ denote the prediction set for account u at parameter q. Using the nonconformity score $s(u, c) = 1 - \widehat{p}_{u,c}$, one possible set construction is

$$\Gamma_q(\mathbf{x}_u) = \{c \in \{0, 1\} : 1 - \widehat{p}_{u,c} \leq q\}. \tag{22}$$

For this construction, increasing q expands the prediction set. Consequently, the account-level

miscoverage loss is non-increasing in q. For a calibration unit g, the corresponding empirical loss is

$$L_g(q) = \frac{1}{|g|} \sum_{u \in g} \mathbb{I}\left[y_u \notin \Gamma_q(\mathbf{x}_u)\right]. \tag{23}$$

The calibration procedure evaluates this loss over candidate values of q and applies the finite-sample correction specified by the selected conformal risk-control algorithm. Under the theorem's applicable assumptions, the target is to control the marginal expected loss for a future exchangeable unit:

$$\mathbb{E}\left[L_{G_{K+1}}(\widehat{q})\right] \leq \alpha, \tag{24}$$

where K is the number of calibration units, G_{K+1} is a future unit, and α is the prespecified target risk. Equation 24 expresses the intended guarantee under the required assumptions; it is not an unconditional guarantee under arbitrary campaign shift. This loss controls marginal prediction-set miscoverage at the campaign-unit level, not error conditional on receiving an automatic singleton label. Selective accuracy is therefore reported separately, and no selective-risk guarantee is claimed unless a calibration method explicitly controls that loss. The number of independent calibration units and the smallest risk level attainable under the selected finite-sample correction are reported; when too few units are available, the calibration result is treated as underpowered or infeasible. The implementation must specify the exact calibration algorithm, candidate grid, correction, and infeasibility handling. If the target cannot be attained, the result is reported as infeasible rather than silently relaxing the target or adjusting it using test labels.

Selective prediction and final decision: After calibration, the selected decision rule is applied unchanged to test accounts. The automatic-decision indicator is defined as

$$A_u = \mathbb{I}\left[|\Gamma_{\widehat{q}}(\mathbf{x}_u)| = 1\right]. \tag{25}$$

When $A_u = 1$, the system emits the unique class in the prediction set as the automatic label. When $A_u = 0$, the system abstains and refers the account for further review. This includes both non-singleton and empty prediction sets; empty sets are counted as abstentions and as miscoverage whenever the true class is excluded. This mechanism distinguishes a unique prediction from an unresolved prediction instead of forcing a label for every account.

The singleton rate measures the proportion of accounts receiving automatic labels:

$$\mathrm{SR} = \frac{1}{N} \sum_{u=1}^{N} A_u, \tag{26}$$

and the abstention rate is

$$\mathrm{AR} = 1 - \mathrm{SR}. \tag{27}$$

Selective accuracy is calculated only among automatically labeled accounts:

$$\mathrm{Acc}_{\mathrm{sel}} = \frac{\sum_{u=1}^{N} A_u \mathbb{I}[\widehat{y}_u = y_u]}{\sum_{u=1}^{N} A_u}, \tag{28}$$

provided that at least one account receives an automatic label. If no account receives an automatic label, selective accuracy is reported as undefined. The singleton rate and abstention

rate must be reported with selective accuracy because a model can increase selective accuracy by abstaining on more difficult accounts.

Performance assessment: The final component evaluates both classification performance and uncertainty-aware decision behavior. Macro-F1 measures class-balanced classification quality, PR-AUC evaluates precision–recall discrimination, and MCC summarizes agreement between predicted and ground-truth classes using all confusion-matrix outcomes. The selective-prediction analysis additionally reports singleton rate, abstention rate, selective accuracy, prediction-set size, empirical coverage, and miscoverage.

Empirical coverage is the fraction of test accounts whose true labels are contained in their prediction sets. It is reported separately from selective accuracy and is not presented as a formal guarantee unless the selected calibration theorem and its assumptions justify that interpretation. Similarly, a guarantee on prediction-set miscoverage does not automatically guarantee low error among singleton predictions. If false-discovery risk among automatic labels is an operational requirement, it must be defined and calibrated as a separate objective.

The experimental evaluation compares the complete model against suitable feature-only and graph-based baselines and performs component ablations under the same data-partitioning protocols. Repeated runs, confidence intervals, and appropriate statistical comparisons are used to quantify uncertainty in the reported results. These experiments determine whether the proposed temporal coordination representation improves generalization and whether selective prediction provides a useful trade-off between automatic labeling and abstention.

Algorithm 1 summarizes the complete execution flow. It partitions the data, constructs account representations and the null-corrected temporal graph, trains the relational predictor, and calibrates the decision rule using separate calibration units. The fixed model and calibrated rule are then applied to the final test accounts, after which automatic predictions, abstentions, and evaluation metrics are generated.

6 Experimental Evaluation

This section defines an experimental protocol to evaluate whether campaign-aware temporal coordination modeling improves social bot detection under distribution shift and whether selective prediction provides more reliable decisions when the test campaign differs from the calibration data. The evaluation is organized around eight objectives: (i) quantify the performance degradation caused by campaign shift; (ii) measure the contribution of temporal coordination to unseen-campaign detection; (iii) evaluate null-corrected co-activity modeling; (iv) assess campaign-unit calibration and prediction-set reliability; (v) characterize the trade-off between automatic prediction and abstention; (vi) test robustness to incomplete or perturbed activity observations; (vii) measure computational scalability; and (viii) evaluate cross-dataset transfer.

The experiments distinguish three outcomes that should not be conflated: classification performance, empirical prediction-set reliability, and computational cost. High classification accuracy does not establish calibration, and empirical coverage on a finite collection of campaigns does not, by itself, establish a distribution- free guarantee for unseen campaigns.

6.1 Research Questions and Objective Alignment

The experiments are designed to answer the following research questions.

Algorithm 1 Campaign-Shift-Aware Social Bot Detection

Require: Account profiles $\mathcal{P}$, events $\mathcal{E}$, labels $\mathcal{Y}$, campaign identifiers $\mathcal{C}$, optional follow graph $\mathcal{G}_f$, cutoff T, target risk α

Ensure: Test predictions, abstention decisions, and evaluation metrics

```
Partition data into training, calibration, and test sets
Enforce campaign separation and temporal cutoffs
Extract profile features and temporal rhythm representations
Fit preprocessing transformations using training data only
Compute multiscale temporal co-activity scores
Estimate the configured time-permutation null distribution
Correct edge weights and sparsify the co-activity graph
Derive coordination statistics and incorporate permitted graph relations
Train the relational predictor using labeled training data
Fix model parameters before calibration
Generate calibration predictions and compute unit-level losses
Apply the selected risk-control algorithm to obtain q̂ or report infeasibility
Generate test predictions using the fixed predictor
Construct test prediction sets using q̂
for all test accounts u do
    if |Γ_q̂(x_u)| = 1 then
        Emit the unique class as an automatic prediction
    else
        Abstain and refer the account for review
    end if
end for
Compute classification and selective-prediction metrics
Compare baselines and ablations under each evaluation protocol
Report statistical uncertainty and calibration assumptions
```

- **RQ1: Campaign-shift sensitivity.** How substantially does social bot detection deteriorate when evaluation changes from random account-level splitting to campaign-disjoint evaluation?
- **RQ2: Temporal coordination effectiveness.** Does temporal coordination modeling improve detection on unseen campaigns beyond account-profile and individual activity-rhythm features?
- **RQ3: Null-corrected coordination.** Does correcting co-activity scores against a suitable null model improve discrimination between coordinated automation and coincidental activity?
- **RQ4: Calibration under campaign shift.** How do account-level, campaign-blocked, and other defensible calibration procedures behave when evaluated on held-out campaigns?
- **RQ5: Selective prediction.** How does prediction reliability change as the system abstains from an increasing proportion of uncertain account predictions?
- **RQ6: Robustness.** How sensitive is the proposed model to missing events, timestamp perturbations, low activity, and legitimate synchronized communities?
- **RQ7: Computational scalability.** How do graph construction, training, inference time, and memory requirements scale with the number of accounts and observed interactions?

Table 2: Alignment between research objectives and experimental evidence.

Objective	Question	Experiment	Required evidence
O1	Quantify campaign-shift degradation	E1	Performance difference between random and campaign-disjoint evaluation
O2	Evaluate detection on unseen campaigns	E2, E4	Unseen-campaign performance and controlled component ablations
O3	Isolate the incremental contribution of null-model correction	E4, E6	Performance and graph-quality changes with and without null correction
O4	Assess campaign-unit calibration	E3	Unit-level miscoverage, calibration-unit count, uncertainty intervals, and explicit assumption checks
O5	Quantify abstention–reliability trade-offs	E3	Coverage, singleton rate, abstention, selective accuracy, and set size
O6	Evaluate robustness to observation changes	E5	Performance and selective-prediction changes under controlled perturbations
O7	Measure scalability	E7	Runtime, peak memory, graph size, and inference throughput
O8	Test transfer across datasets	E8	Source-to-target performance and calibration degradation

- **RQ8: Cross-dataset generalization.** Does the model retain useful detection performance when transferred between datasets collected under different sampling, annotation, and platform conditions?

Table 2 maps these questions to the experiments and the evidence required to answer them.

6.2 Datasets and Data Audit

The evaluation will use publicly documented social bot detection datasets, subject to the availability of legally accessible data, reproducible account identifiers, relevant temporal observations, and sufficient label provenance. Candidate datasets include Cresci-2017, Bot Repository collections, TwiBot-20, and, where access and compatibility permit, TwiBot-22. These datasets are candidates for evaluation rather than an assertion that every dataset contains verified campaign annotations or all fields required by the proposed method.

Table 3 specifies the information that must be verified before each dataset is included in the final evaluation. The values should be populated from the actual dataset files and their accompanying documentation.

For each included dataset, the audit will record the number of accounts, human and bot labels, available events, timestamp coverage, profile-feature completeness, graph size, duplicate rate, missing values, and number of independent campaign or collection units. The audit will also document how labels were assigned and whether the collection procedure could introduce artifacts correlated with the target label.

An existing dataset grouping will be treated as a campaign unit only when its provenance supports that interpretation. Collection batches, account creation periods, profile templates, or bursts of activity may be used to construct proxy groups for sensitivity analysis, but such groups will not be described as verified operational campaigns. Experiments using proxy groups will be explicitly distinguished from those using independently established campaign annotations.

Table 3: Dataset audit and eligibility requirements.

Dataset	Intended role	Required audit
Cresci-2017	Candidate source dataset	Verify account labels, collection-group provenance, available activity fields, timestamps, duplicate accounts, and the number of independent groups.
Bot Repository collections	Candidate source or auxiliary dataset	Verify the exact collection version, label definitions, collection period, account overlap, and compatibility of available fields.
TwiBot-20	Candidate source or target dataset	Verify account-level labels, profile and relation availability, temporal coverage, collection procedures, and possible overlap with other datasets.
TwiBot-22	Optional extended-scale or transfer dataset	Verify access conditions, label provenance, timestamp availability, graph construction requirements, and computational feasibility.

The final manuscript will report the exact dataset version, source, access date where applicable, preprocessing configuration, and resulting sample counts. No dataset will be included in a comparison unless the input features and evaluation labels required for that comparison are available and compatible.

6.3 Data Preprocessing and Leakage Prevention

Preprocessing is performed separately for each experimental split to prevent information from the calibration or test campaigns from influencing model development.

Account identifiers are normalized to a consistent representation, and duplicate records are resolved using documented rules. Timestamps are converted to a consistent time representation, and events with invalid or unusable timestamps are handled according to a prespecified policy. Missing profile attributes are imputed using statistics estimated from the training partition only. Categorical variables are encoded using a training-fitted mapping, and numerical features are standardized using training-partition statistics.

The following controls are applied throughout the experiments.

- **Duplicate control:** duplicate accounts and repeated records are identified before splitting. Where cross-dataset account identity can be established reliably, overlapping accounts are either removed from the relevant transfer evaluation or explicitly handled by a predefined overlap policy.
- **Temporal control:** a feature for an account at prediction time t may use only information available at or before t. Future interactions, future profile changes, and post-cutoff graph edges are excluded from historical predictions.
- **Partition control:** all accounts belonging to a held-out campaign remain outside predictor training and hyperparameter selection. Calibration and final test campaigns are also separated according to the selected calibration protocol.
- **Graph control:** graph construction, edge filtering, feature normalization, and any data-dependent graph thresholds are fitted or selected using permitted training data only. Test-account edges must not be used to construct training representations if doing so exposes held-out interactions or labels.

- **Label control:** test labels are unavailable during feature design, model selection, threshold selection, and calibration. They are used only for the final evaluation.
- **Tuning control:** all model-selection decisions are made using training and validation partitions. The test campaigns are not repeatedly inspected to choose hyperparameters or reporting thresholds.

When a graph model requires relations involving previously unseen accounts, the evaluation will specify whether inference is inductive or transductive. In particular, access to unlabeled test topology will be reported separately from access to test labels. The same information-access rules will be applied to all competing methods whenever their architectures permit a comparable evaluation.

6.4 Evaluation Protocols

A central requirement of this study is to distinguish ordinary generalization from campaign, temporal, and dataset shift. These settings answer different scientific questions and will not be treated as interchangeable.

Let $\mathcal{C}$ denote the set of eligible campaign or collection units. For campaign-aware experiments, the partition is performed over units rather than independently over accounts.

P0: Random account-level split. Accounts are randomly divided into training, validation, calibration, and test partitions, subject to the dataset's available grouping information. This protocol serves as a conventional reference condition. If accounts from the same campaign occur in multiple partitions, the resulting performance is not interpreted as evidence of unseen-campaign generalization.

P1: Campaign-disjoint split. Complete campaigns or eligible collection units are assigned to training, validation, calibration, and test partitions. No account from a held-out test campaign is used to train the predictor or select its hyperparameters. This protocol is the primary test of campaign-disjoint generalization.

P2: Multi-campaign holdout. Multiple campaigns are held out together to evaluate whether performance depends on a small number of relatively easy target campaigns. Where the dataset contains enough independent units, campaign holdouts are rotated across repeated group-level splits. Each split records the number of training, calibration, and test campaigns.

P3: Chronological evaluation. A prespecified time cutoff separates historical training observations from later observations. All features for a prediction are computed using information available by its prediction time. When campaigns span the cutoff, the protocol reports this fact and does not automatically label the setting campaign-disjoint. Chronological generalization tests future-time shift, which is distinct from campaign-disjoint generalization.

P4: Cross-dataset transfer. The model is trained on a source dataset and evaluated on a separate target dataset. Source and target features are mapped to a documented common representation. Features unavailable in the target dataset are not reconstructed using target

Table 4: Evaluation protocols and their intended interpretation.

Protocol	Partition principle	Primary interpretation
P0	Random account-level partition	Conventional reference performance
P1	Disjoint campaigns or collection units	Unseen-campaign generalization
P2	Multiple held-out campaigns	Sensitivity to target-campaign composition
P3	Historical versus future observations	Temporal generalization
P4	Source versus target dataset	Cross-dataset transfer

labels or hidden annotations. Any target-side unsupervised adaptation is reported as a separate setting and is not described as zero-shot transfer.

Table 4 summarizes the role of each protocol.

All split manifests will identify the included accounts or units, the split seed, temporal cutoff where applicable, and the grouping rule. The number of independent units will be reported alongside the number of accounts because thousands of accounts drawn from a small number of campaigns do not provide thousands of independent campaign-level observations.

6.5 Compared Methods and Baselines

The proposed method will be compared with methods representing different feature, relational, temporal, and uncertainty-modeling strategies. Baseline selection will depend on verified implementation availability and input compatibility.

Conventional account-level baselines. A support vector machine and a random forest will be trained using available profile and individual activity features. These baselines establish whether the proposed relational representation improves upon conventional account-level classification.

Profile and rhythm baseline. A model using account-profile features and individual activity-rhythm features, but no co-activity graph, will isolate the contribution of coordination modeling. Its feature budget and classifier capacity will be kept comparable to the corresponding components of the proposed system where practical.

Graph-learning baselines. GraphSAGE and graph attention networks (GAT) will be included when their input graphs and inductive or transductive assumptions are compatible with the evaluation. BotRGCN will be considered when the required relation types and features are available. Temporal graph methods such as TGN or TGAT will be included only when timestamped interactions satisfy their input requirements.

Coordination and temporal baselines. The proposed model will be compared with a fixed-window binary co-activity graph and a timestamp-aware graph without null correction. Where a published coordination-based detector has a verifiable implementation and compatible inputs, it may be included as an additional baseline. Any reimplementation will be identified as such, with the code source, configuration, and deviations documented.

Calibration and selective-prediction baselines. The classification model will be evaluated using raw predicted probabilities and a fixed decision threshold, temperature scaling, and an account-level split-conformal baseline where its assumptions are appropriate. Campaign-blocked calibration will be included when independent calibration campaigns are available. A shift-aware

Table 5: Comparison methods and the factors they isolate.

Method	Input or mechanism	Purpose
SVM / random forest	Profile and individual activity features	Conventional classification reference
Profile–rhythm model	Profile attributes and temporal activity summaries	Contribution of individual temporal behavior
GraphSAGE / GAT	Compatible account-relational graph	Contribution of relational learning
Compatible temporal GNN	Timestamped interactions or temporal graph events	Contribution of temporal graph learning
Fixed-window co-activity	Binary or fixed-window co-activity edges	Reference for temporal coordination
Group-DRO / group reweighting	Group-aware training objective or sample weights, when meaningful training-group labels are available	Tests whether group-robust training explains gains independently of the proposed graph representation
Domain-generalization baseline	A reproducible established method selected before final testing; exact implementation and configuration reported	Reference for generalization across training groups
Timestamp-aware graph without null correction	Temporal edges without null-model adjustment	Contribution of null correction
Proposed model	Profile, rhythm, null-corrected temporal coordination, and relational learning	Combined representation under evaluation
Raw probabilities / temperature scaling	Classifier scores with standard decision or scaling rules	Probability calibration reference
Account-level split conformal	Account-level calibration units	Calibration reference under its applicable assumptions
Campaign-blocked calibration / risk control	Campaign-level calibration units and a specified procedure	Assessment of campaign-unit uncertainty control

or risk-control method will be included only when its algorithm, assumptions, and guarantees can be stated precisely and implemented faithfully.

Campaign-blocked calibration and campaign-level risk control are not assumed to be equivalent. Blocking observations by campaign changes the calibration sampling unit, but it does not automatically provide a finite-sample guarantee under arbitrary distribution shift.

Table 5 lists the principal comparison methods.

All baselines will use the same eligible data partitions and comparable hyperparameter-search budgets. A method will be marked as unavailable rather than silently replaced if its required features, source code, or implementation details cannot be verified.

6.6 Implementation and Hyperparameter Selection

The implementation will record the software versions, hardware configuration, random seeds, model parameter counts, graph construction settings, and training duration. The final manuscript will report the actual environment used rather than assuming a particular hardware platform.

The main hyperparameters include the hidden dimension, learning rate, batch size, number of training epochs, dropout rate, number of relational message-passing layers, number of temporal scales, temporal decay parameters, edge-pruning threshold, and null-model configuration. Candidate search ranges will be established before final testing and reported in a supplementary table. The selected values will be determined using validation data only.

For the parameterized temporal kernel, the weights and time scales will satisfy

$$\pi_m \geq 0, \qquad \sum_{m=1}^{M} \pi_m = 1, \qquad \lambda_m > 0. \tag{29}$$

The search procedure will enforce these constraints, for example through normalized nonnegative candidate weights and positive candidate time scales. Unless a differentiable graph-construction method is implemented, these parameters will not be described as learned end-to-end through the GNN.

The null-model procedure will specify the randomization scheme, number of randomizations or analytical approximation, and the treatment of sparse accounts. Null-model parameters will be estimated without using test labels. Any graph-pruning threshold selected empirically will be selected on the validation partition and frozen before final evaluation.

Each configuration will be trained with multiple independent random seeds. Hyperparameter selection will not use final test performance. The final report will provide the selected configuration, the search space, stopping rule, and whether the results are based on a single trained model or an ensemble.

6.7 Evaluation Metrics

The metrics are divided into classification quality, probability calibration, selective prediction, campaign-level reliability, and computational efficiency. Each metric is computed on the same eligible test partition for all methods being compared.

Classification performance. Let TP, TN, FP, and FN denote the true-positive, true-negative, false-positive, and false-negative counts, respectively. Macro-F1 is calculated as the unweighted mean of the class-specific F1 scores:

$$\text{MacroF1} = \frac{1}{2}\left(\frac{2TP}{2TP + FP + FN} + \frac{2TN}{2TN + FP + FN}\right). \tag{30}$$

The implementation will use the equivalent class-wise definition with a documented zero-division convention for degenerate cases.

Precision–recall area (PR-AUC), ROC-AUC, and the Matthews correlation coefficient (MCC) will also be reported. PR-AUC is particularly important when the bot and human classes are imbalanced. The false positive rate at a prespecified true positive rate will be reported where the score distribution supports the requested operating point. Thresholds will be selected on validation data, not the test set.

Probability calibration. The Brier score evaluates the squared error between predicted probabilities and one-hot labels:

$$\text{Brier} = \frac{1}{N}\sum_{u=1}^{N}\sum_{c\in\{0,1\}} \left(\hat{p}_{u,c} - \mathbb{I}[y_u = c]\right)^2. \tag{31}$$

Expected calibration error (ECE) will be computed using a fixed, documented binning rule:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left|\text{acc}(I_b) - \text{conf}(I_b)\right|, \tag{32}$$

where I_b is the set of test observations in bin b. The number of bins and binning strategy will be fixed before test evaluation. ECE will be interpreted as a descriptive calibration metric rather than a formal guarantee.

Prediction-set quality. For a prediction set $\Gamma(\mathbf{x}_u)$, empirical marginal coverage is

$$\widehat{\mathrm{Cov}} = \frac{1}{N}\sum_{u=1}^{N} \mathbb{I}\left[y_u \in \Gamma(\mathbf{x}_u)\right]. \tag{33}$$

Mean prediction-set size is

$$\overline{S} = \frac{1}{N}\sum_{u=1}^{N} |\Gamma(\mathbf{x}_u)|. \tag{34}$$

Coverage and set size will be reported together because a method can obtain high coverage simply by returning large sets. For binary classification, a set containing both labels is an ambiguous result, whereas a singleton set supports an automatic class decision under the specified decision policy.

Selective prediction. Let

$$A_u = \mathbb{I}\left[|\Gamma(\mathbf{x}_u)| = 1\right] \tag{35}$$

denote an automatic-decision indicator. The singleton rate and abstention rate are

$$\mathrm{SR} = \frac{1}{N}\sum_{u=1}^{N} A_u, \qquad \mathrm{AR} = 1 - \mathrm{SR}\,. \tag{36}$$

Selective accuracy is defined only when at least one singleton prediction is produced:

$$\mathrm{Acc}_{\mathrm{sel}} = \frac{\sum_{u=1}^{N} A_u\, \mathbb{I}[\hat{y}_u = y_u]}{\sum_{u=1}^{N} A_u}. \tag{37}$$

Selective Macro-F1 will also be computed over the automatically classified accounts, using a consistent class-label and zero-division convention. Results will report the fraction of accounts receiving automatic decisions so that selective accuracy cannot be interpreted without its associated coverage.

Campaign-unit reliability. Let $\mathcal{G}_{\mathrm{test}}$ denote the independent test campaigns, and let G be one such campaign. Its empirical miscoverage is

$$\widehat{L}_G = \frac{1}{|G|}\sum_{u\in G} \mathbb{I}\left[y_u \notin \Gamma(\mathbf{x}_u)\right]. \tag{38}$$

The mean campaign-level miscoverage is

$$\overline{L}_{\mathrm{campaign}} = \frac{1}{|\mathcal{G}_{\mathrm{test}}|}\sum_{G\in\mathcal{G}_{\mathrm{test}}} \widehat{L}_G. \tag{39}$$

This aggregate gives each evaluated campaign equal weight; the number of independent campaigns and their sizes are reported alongside it.

6.8 Statistical Analysis and Falsification Criteria

The following decision rules are specified before examining final test results. They are analysis criteria, not claims that the hypotheses have already been supported. Let $M_{m,p}$ denote a metric for method m under protocol p.

H1: Random-split optimism. Define $\Delta_{\text{shift}} = M_{\text{proposed},P1} - M_{\text{proposed},P0}$ for Macro-F1 and PR-AUC, where P0 is the random account split and P1 is the campaign- or eligible-group-disjoint split. The direction and magnitude of the observed difference will be reported with a 95% confidence interval. Evidence for a degradation claim requires the interval for Δ_{shift} to lie below zero; otherwise, the claim is not supported. If the group labels are only collection proxies, the conclusion is limited to collection-group shift.

H2: Incremental coordination value. For the proposed model versus its strongest prespecified non-coordination baseline, define $\Delta_{\text{coord}} = M_{\text{proposed},P1} - M_{\text{baseline},P1}$. An improvement claim requires a positive 95% confidence interval for the paired difference on the primary metric, Macro-F1, with PR-AUC reported as a complementary metric. The null-correction contribution is assessed separately by comparing the same architecture with and without null correction.

Unit of inference and tests. Where there are enough independent groups, paired differences are computed on the same held-out groups. The primary inferential procedure is a paired permutation test over group-level differences; a Wilcoxon signed-rank test is reported as a sensitivity analysis when its assumptions are tenable. Confidence intervals are obtained by resampling independent groups rather than individual accounts. With very few groups, all estimates are labeled low-power/descriptive and exact attainable p-value resolution is reported; account-level resampling is not used to manufacture a large effective sample size. For random-split repeats, seed-level results are summarized separately and are not treated as independent campaigns.

Multiplicity. Macro-F1 under P1 is the primary endpoint for the principal proposed-model comparison. Other protocol–metric–baseline comparisons are secondary. Within each prespecified family of secondary hypotheses, p-values are adjusted using the Holm procedure. Both adjusted and unadjusted values, effect sizes, confidence intervals, number of groups, and group sizes are reported. Missing or undefined metric values are not imputed as favorable results.

Calibration and class-conditional reliability. Marginal coverage and campaign-level miscoverage are reported alongside class-conditional coverage for the human and bot classes, where sample sizes permit. A Mondrian/class-conditional calibration variant may be included as a separate method, but it is not assumed to provide a useful finite-sample guarantee when class-specific calibration counts are small. For each calibration condition, the number of independent units K and the minimum attainable risk level implied by the implemented finite-sample correction are reported. If the target α is unattainable for the observed K, the formal risk-control result is marked infeasible and the empirical calibration analysis is retained without a guarantee.

7 Conclusion

This paper presents a campaign-shift-aware framework for social bot detection that integrates temporal coordination modeling, null-model correction, graph neural networks, and uncertainty-aware selective prediction. The proposed evaluation methodology examines generalization to unseen campaigns, prediction reliability, and the trade-off between automatic classification and human review. Campaign-disjoint, chronological, and cross-dataset experiments, together with controlled ablations, are required to establish the contribution of each component. The present manuscript defines a testable methodology; it does not claim empirical superiority before the experiments are executed. Campaign-level risk control is treated as empirical unless the calibration-unit assumptions and finite-sample requirements are verified. The next step is to execute the P0–P1 comparison and core ablations on audited datasets, report independent-group counts and uncertainty, and evaluate robustness and scalability.

References

[1] S. Cresci, "A decade of social bot detection," *Communications of the ACM*, vol. 63, no. 10, pp. 72–83, 2020.

[2] S. Feng, H. Wan, N. Wang, J. Li, and M. Luo, "TwiBot-20: A comprehensive Twitter bot detection benchmark," in *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021, pp. 4485–4494, doi: 10.1145/3459637.3482019.

[3] S. Feng, H. Wan, N. Wang, J. Li, and M. Luo, "SATAR: A self-supervised approach to Twitter account representation learning and its application in bot detection," in *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021.

[4] N. Chavoshi, H. Hamooni, and A. Mueen, "On-demand bot detection and archival system," in *Proceedings of the 26th International Conference on World Wide Web Companion*, 2017, pp. 183–187, doi: 10.1145/3041021.3054733.

[5] T. Khaund, B. Kirdemir, N. Agarwal, H. Liu, and F. Morstatter, "Social bots and their coordination during online campaigns: A survey," *IEEE Transactions on Computational Social Systems*, vol. 9, no. 2, pp. 530–545, 2022, doi: 10.1109/TCSS.2021.3103515.

[6] S. Feng, H. Wan, N. Wang, and M. Luo, "BotRGCN: Twitter bot detection with relational graph convolutional networks," in *Proceedings of the 2021 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 2021.

[7] S. Feng, Z. Tan, R. Li, and M. Luo, "Heterogeneity-aware Twitter bot detection with relational graph transformers," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 4, pp. 3977–3985, 2022, doi: 10.1609/aaai.v36i4.20314.

[8] S. Feng et al., "TwiBot-22: Towards graph-based Twitter bot detection," in *Advances in Neural Information Processing Systems*, vol. 35, Datasets and Benchmarks Track, 2022.

[9] B. He, Y. Yang, Q. Wu, H. Liu, R. Yang, H. Peng, X. Wang, Y. Liao, and P. Zhou, "Dynamicity-aware social bot detection with dynamic graph transformers," in *Proceedings of*

the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI), 2024, pp. 5844–5852, doi: 10.24963/ijcai.2024/646.

[10] F. Liu and R. Ma, “From the past to the present: A social bot detection method based on spatio-temporal interactive perception,” *Knowledge-Based Systems*, vol. 322, Art. no. 113712, 2025, doi: 10.1016/j.knosys.2025.113712.

[11] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[12] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, “Conformal risk control,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

[13] D. Pacheco, A. Hui, C. Torres-Lugo, B. Truong, A. Flammini, and F. Menczer, "Uncovering coordinated networks on social media: Methods and case studies," in *Proceedings of the International AAAI Conference on Web and Social Media*, 2021.

[14] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, "Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization," in *International Conference on Learning Representations*, 2020.

[15] I. Gulrajani and D. Lopez-Paz, "In search of lost domain generalization," in *International Conference on Learning Representations*, 2021.

[16] R. J. Tibshirani, R. Foygel Barber, E. Candes, and A. Ramdas, "Conformal prediction under covariate shift," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[17] R. F. Barber, E. J. Candes, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *The Annals of Statistics*, vol. 51, no. 2, pp. 816–845, 2023.